

CESPPL FIELDVISION PLATFORM

Final Project Summary

Executive Summary

The CESPPL FieldVision Platform is an AI-powered web application developed to automate the classification and management of municipal field-operation images. The project addresses the challenge of manually sorting and monitoring large numbers of operational photographs collected during sanitation and maintenance activities.

A dataset containing 4,179 raw field images across ten operational classes was prepared through validation, preprocessing, duplicate removal, and stratified splitting. After identical-image deduplication, 3,616 processed images were used for model development. An EfficientNetB0 transfer-learning model was trained and achieved a final test accuracy of **95.03%**, a **macro F1-score of 93.20%**, and a best validation accuracy of **95.58%**.

The trained model was integrated into a Streamlit-based platform with image upload, AI prediction, SQLite storage, dashboard analytics, image browsing, and report-generation capabilities, providing an efficient decision-support system for municipal field operations.

Problem Statement

Municipal field teams capture thousands of photographs representing activities such as road sweeping, bin washing, beach cleaning, and waste collection. Manually organizing and reviewing these images is time-consuming, inconsistent, and difficult to scale.

The objective of this project is to automatically classify uploaded field-operation images using deep learning while providing an easy-to-use dashboard for monitoring uploads, reviewing predictions, browsing records, and generating reports.

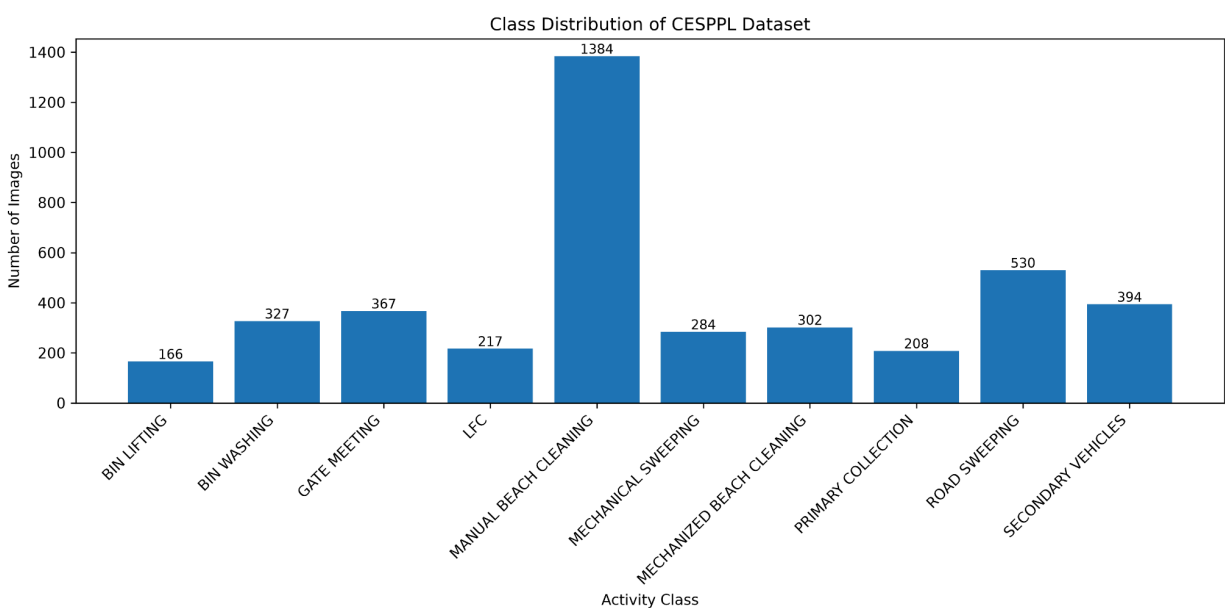
Dataset

The dataset consists of real municipal field-operation photographs collected across ten activity classes.

Dataset Summary

- Raw Images: ****4,179****
- Processed Images After Deduplication: ****3,616****
- Number of Classes: ****10****
- Train/Validation/Test Split: ****70% / 15% / 15%****

Class Distribution



The dataset is naturally imbalanced. MANUAL BEACH CLEANING is the largest class, while BIN LIFTING and PRIMARY COLLECTION contain fewer samples.

Methodology

The project followed a complete machine-learning workflow:

- Dataset inventory and validation
- EXIF correction
- Image resizing and centre cropping
- RGB conversion
- Duplicate-image removal

- Stratified train/validation/test split
- Transfer learning using EfficientNetB0
- Model evaluation and final model selection
- Streamlit deployment with SQLite database integration

Results

Overall Performance

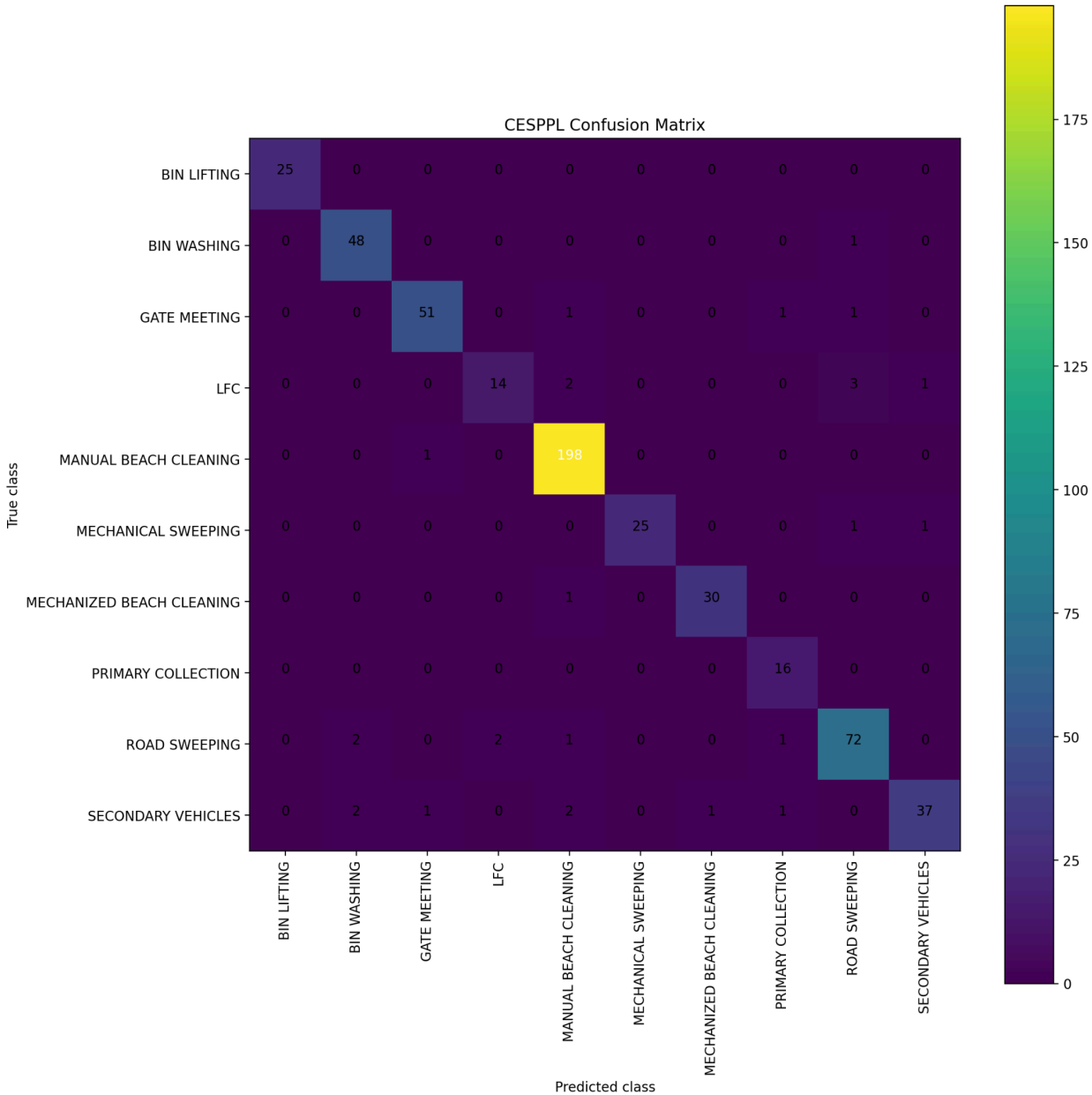
Metric	Value
Best Validation Accuracy	95.58%
Final Test Accuracy	95.03%
Macro F1-Score	93.20%

Per-Class Performance

Class	Precision	Recall	F1-score	Support
BIN LIFTING	1.0000	1.0000	1.0000	25
BIN WASHING	0.9231	0.9796	0.9505	49
GATE MEETING	0.9623	0.9444	0.9533	54
LFC	0.8750	0.7000	0.7778	20
MANUAL BEACH CLEANING	0.9659	0.9950	0.9802	199
MECHANICAL SWEEPING	1.0000	0.9259	0.9615	27
MECHANIZED BEACH CLEANING	0.9677	0.9677	0.9677	31
PRIMARY COLLECTION	0.8421	1.0000	0.9143	16

ROAD SWEEPING	0.9231	0.9231	0.9231	78
SECONDARY VEHICLES	0.9487	08409	0.8916	44

Confusion Highlights



The confusion matrix indicates excellent performance across most classes. The most challenging activities were visually similar operations, including:

- LFC and ROAD SWEEPING
- LFC and MANUAL BEACH CLEANING
- ROAD SWEEPING and BIN WASHING
- ROAD SWEEPING and LFC
- SECONDARY VEHICLES and BIN WASHING
- SECONDARY VEHICLES and MANUAL BEACH CLEANING

These results suggest that similar backgrounds, machinery, and operational contexts can make classification more difficult.

Error Analysis

The remaining prediction errors mainly occurred due to:

- Similar-looking operational activities
- Images captured from different distances
- Partial visibility of equipment
- Multiple activities appearing in a single image
- Class imbalance
- Similar vehicles and road environments

Increasing the diversity and quantity of training images for underrepresented classes could further improve performance.

System Implementation

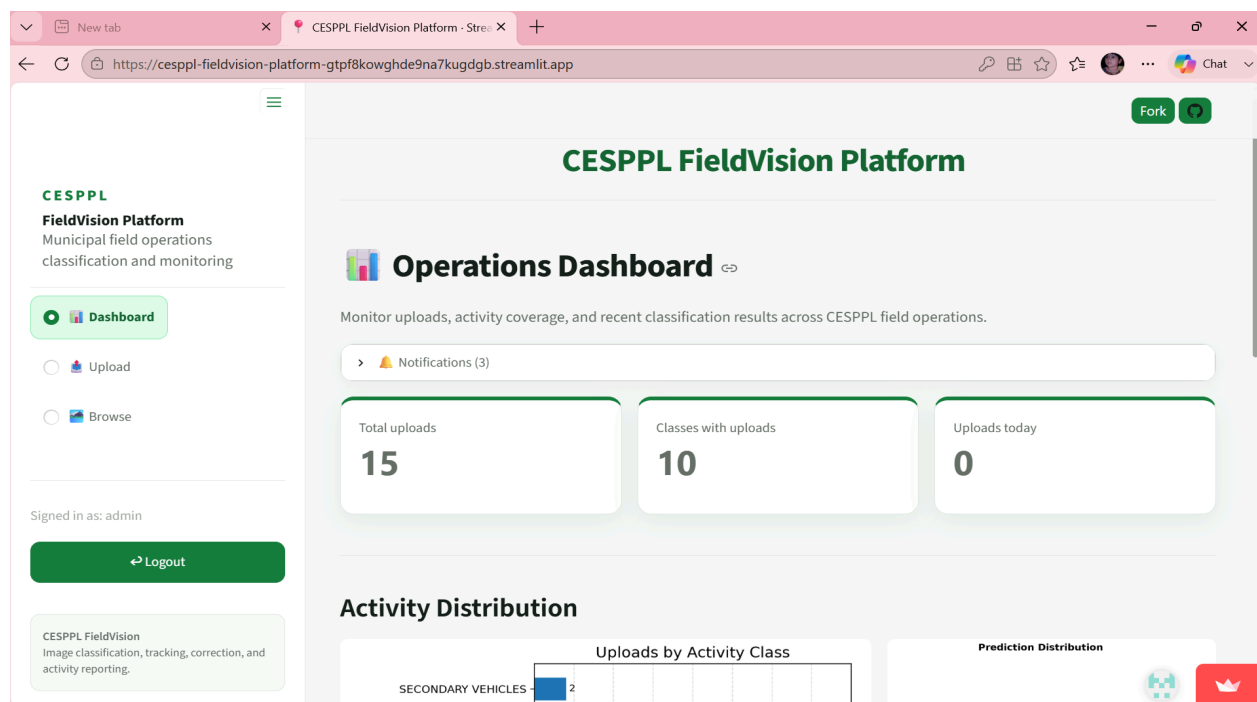
The CESPPL FieldVision Platform was developed as a complete AI-powered web application using Streamlit for the frontend and TensorFlow/Keras for deep learning inference.

The system workflow consists of:

- Image upload through the Streamlit interface
- Image preprocessing (RGB conversion, resizing, normalization)
- Prediction using the trained EfficientNetB0 model

- Storage of uploaded images and prediction records in a SQLite database
 - Dashboard visualization of operational statistics
 - Image browsing and search functionality
 - Class reassignment for correcting predictions
 - ZIP download support for reports and records
- The platform provides administrators with a centralized dashboard for monitoring municipal field-operation activities.

Dashboard



Limitations

Although the platform achieved strong performance, several limitations remain:

- The dataset is naturally imbalanced, with some activities having significantly fewer samples.
- Certain activities share similar visual characteristics, making classification more challenging.
- The model has been evaluated only on the available CESPPL dataset and may require retraining for other organizations or environments.
- The current authentication system supports a single administrator.
- SQLite is appropriate for small-scale deployments but is not intended for large multi-user

systems.

- When deployed on Streamlit Community Cloud, locally stored database records may not persist after application restarts.
- Predictions should support human decision-making and not replace manual verification for important operational records.

Future Work

Potential improvements for future versions include:

- Retraining the model using corrected or reassigned uploads.
- Expanding the dataset with additional field-operation images.
- Implementing secure authentication with multiple user roles.
- Migrating from SQLite to a persistent cloud database.
- Adding continuous model monitoring and performance tracking.
- Improving the mobile experience for field personnel.
- Exploring advanced deep-learning architectures for even higher classification accuracy.

Conclusion

The CESPPL FieldVision Platform demonstrates a complete end-to-end artificial intelligence workflow, beginning with raw image inventory and progressing through preprocessing, duplicate removal, model training, evaluation, deployment, and monitoring.

Using transfer learning with EfficientNetB0, the system achieved a final test accuracy of ****95.03%**** and a ****macro F1-score of 93.20%**** across ten municipal field-operation classes.

The Streamlit application successfully integrates AI-powered image classification with dashboard analytics, image management, and operational reporting.

The project provides an effective decision-support tool for municipal field operations while also serving as a strong foundation for future enhancements such as cloud-based storage, multi-user support, and continuous model improvement.

